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10 CFR 50.73

NLS2024075
December 31, 2024

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Subject: Licensee Event Report No. 2024-007-00
Cooper Nuclear Station, Docket No. 50-298, Renewed License No. DPR-46

The purpose of this correspondence is to forward Licensee Event Report 2024-007-00.

This letter does not contain regulatory commitments.

Sincerely,

Khalil Dia
Site Vice President

/jo

Attachment: Licensee Event Report 2024-007-00

cc: Regional Administrator w/attachment
USNRC - Region IV

Cooper Project Manager w/attachment
USNRC - NRR Plant Licensing Branch IV

Senior Resident Inspector w/attachment
USNRC - CNS

SRAB Administrator w/attachment

NPG Distribution w/attachment

INPO Records Center w/attachment
via IRIS entry

SORC Chairman w/attachment

CNS Records w/attachment

COOPER NUCLEAR STATION
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NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 04/30/2027
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name		2. Docket Number	3. Page
Cooper Nuclear Station	☒ 050 / ☐ 052	298	1 of 3

4. Title
Inoperable Pressure Switch During Mode Change Results in Operation or Condition Prohibited by Technical Specifications

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
11	03	2024	2024	007	00	12	31	2024	Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
1	018

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Linda Dewhirst, Regulatory Affairs and Compliance Manager	(402) 825-5416

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	SB	PS	P381	N					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On November 3, 2024, with the Cooper Nuclear Station (CNS) reactor in Mode 1, Power Operation, scheduled surveillance testing of the eight Safety Relief Valves (SRV) was being performed. During this testing, pressure switch (MS-PS-300B) did not actuate when SRV71B was taken to the OPEN position. As such, Operations personnel declared MS-PS-300B inoperable.

Review of a picture taken in the area of SRV71B on October 16, 2024, while the reactor was in Mode 5, Refueling, revealed that the flex hose that extends from the SRV tailpipe to the pressure switch may be damaged. This was later confirmed when maintenance personnel found the flex hose on top of the SRV tailpipe and the connection port for MS-PS-300B damaged. Repairs were made and subsequent testing performed satisfactorily, and Operations personnel declared MS-PS-300B operable.

CNS made the change from Mode 4 to Mode 2 while this condition existed, violating Technical Speciation (TS) Limiting Condition for Operation 3.0.4, and is reporting this event as an operation or condition prohibited by plant TS.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Cooper Nuclear Station	05000-298	2024	007	00

NARRATIVE

PLANT STATUS

Cooper Nuclear Station (CNS) was in Mode 1, Power Operation, at 18 percent power at the time of the event on November 3, 2024.

BACKGROUND

The pressure relief system includes three American Society of Mechanical Engineers (ASME) code safety valves (SV) [EIIS: SB] and eight safety relief valves (SRV) [EIIS: RV], all of which are located on the main steam lines [EIIS: SB] within the drywell [EIIS: NH], between the reactor vessel [EIIS: RPV] and the first main steam isolation valve [EIIS: ISV]. The SVs provide protection against over pressurization of the nuclear system and discharge directly into the interior space of the drywell. The SRVs discharge to the suppression pool and provide three main functions: overpressure relief operation to limit the pressure rise and prevent safety valve opening, overpressure safety operation to prevent nuclear system over pressurization, and depressurization operation (opened automatically or manually) as part of the emergency core cooling system [EIIS: BJ, BM, BO].

Each SRV is equipped with a pressure switch mounted in its discharge tailpipe. When a SRVs associated discharge pressure switch senses tailpipe pressure of at least 30 psig, the SRV can be determined to be in the OPEN position. When the pressure switch is actuated at 30 psig, an amber indicating light will illuminate in the Control Room and the annunciator for the relief valve will alarm, indicating the SRV is open.

EVENT DESCRIPTION

On November 3, 2024, while performing scheduled surveillance testing of the eight Safety Relief Valves (SRV) during plant start-up activities from Refueling Outage (RE33), pressure switch (MS-PS-300B) did not actuate when SRV71B was taken to the OPEN position. SRV71B tailpipe temperature did immediately rise, however, the temperature was approximately 100 degrees Fahrenheit lower than expected for a fully open SRV and the temperature profile exhibited erratic behavior. The main turbine bypass valve position responded as expected, confirming SRV71B did fully open. As such, MS-PS-300A was declared inoperable on November 3, 2024.

As a result of the low temperatures on the SRV71B tailpipe, questions arose regarding the possibility of a disconnected flex hose that extends from the SRV71B tailpipe to MS-PS-300B. SRV71B was not cycled open prior to RE33 and was not subjected to abnormal operating conditions that would have induced a vibration or fatigue failure of the flex hose. A picture taken near the area of SRV71B on October 16, 2024, was reviewed and it was discovered that the flex hose may have been inadvertently disconnected.

Maintenance personnel were dispatched to SRV71B and confirmed that the flex hose was disconnected and was laying on top of the SRV71B tailpipe. The threaded connections of the flex hose and the connection port on the SRV71B tailpipe were found damaged.

MS-PS-300B was repaired, surveillance testing was performed satisfactorily, and MS-PS-300B declared operational on November 4, 2024.

BASIS FOR REPORT

The damaged flex hose existed prior to the change from Mode 4, Cold Shutdown, to Mode 2, Startup, as evidenced by the picture taken on October 16, 2024. As such, CNS is reporting this event as an operation or condition which was prohibited by plant TS per 10 CFR 50.73(a)(2)(i)(B). CNS made the change from Mode 4 to Mode 2 while this condition existed, violating Technical Speciation (TS) Limiting Condition for Operation 3.0.4, and is reporting this event as an operation or condition prohibited by plant TS.

SAFETY SIGNIFICANCE

Each SRV is equipped with a pressure switch mounted on its discharge tailpipe. Each SRV can be determined to be open (in part) when its associated discharge pressure switch senses tailpipe pressure of at least 30 psig. Following pressure switch actuation (30 psig), an amber indicating light will illuminate and an annunciator will alarm, indicating the SRV has opened. Although this condition prevented the Control Room to receive the OPEN indication, it would not have prevented the ability to open SRV71B to perform its safety function to prevent reactor vessel over pressurization. This is considered an isolated incident; the station risk of recurrence of the equipment failure is minimal.

CAUSES

The direct cause of the failed SRV surveillance test was caused by a damaged and separated flex hose extending from the SRV71B tailpipe to MS-PS-300B.

The cause, date, and time of the flex hose becoming damaged and separated from SRV71B is unknown, as there is no documentation of potential damage to the equipment in the area. Historical data revealed that surveillance testing performed in November 2022 following RE32 demonstrated that MS-PS-300B was not damaged during RE32 and was operable during Operating Cycle 33. Additionally, SRV71B was not cycled OPEN during Operating Cycle 33, eliminating the possibility that the flex hose damaged occurred prior to RE33.

CORRECTIVE ACTIONS

The flex hose and SRV71B tailpipe connection port were repaired, and subsequent surveillance testing performed satisfactorily. An extent of condition review was performed with no additional common mode failure implications identified.

PREVIOUS EVENTS

A review of CNS licensee event reports for the last three years was performed. There were no previous occurrences involving a similar failure to that discussed in this report.